SchoolQ

Queue Management System

Al-Noor International School

Full Changelog

Detailed Version History & Release Notes

Developer	Vyra Systems
Contact	asserhegazy@vyra-systems.com
Product	SchoolQ — Queue Management System
Client	Al-Noor International School
Generated	14 October 2026
Versions	v1.2.0 !' v1.10.0 (37 releases)

Version Index

v1.2.0	Initial Release	06 May 2026
v1.3.0	Bilingual TTS Fix & Voice Quality	06 May 2026
v1.3.1	Security Hardening & Performance	07 May 2026
v1.3.x	Bug Fix Series (5 patches)	07 May 2026
v1.4.0	Permissions, Logo & Audit Log	07 May 2026
v1.5.15	Service Types Deep Integration	15 May 2026
v1.5.16	Force Logout & Socket Security	20 May 2026
v1.5.17	Input Validation & Security Phases 1–4	20 May 2026
v1.5.18	UX, Migration Runner & Test Suite	20 May 2026
v1.5.25	Cancel Action, Permissions & Ticket Log	02 Jun 2026
v1.5.26	Reset Ticket Data	03 Jun 2026
v1.5.27	Self-Service Kiosk for Parents	03 Jun 2026
v1.5.28	Graceful Port Conflict Handling	03 Jun 2026
v1.5.29	TDZ Crash Fix	03 Jun 2026
v1.5.30	Kiosk Auto-Print & Permission	03 Jun 2026
v1.5.31	Export/Import Users & Settings	03 Jun 2026
v1.5.32	Datatype Mismatch Fix	03 Jun 2026
v1.5.33	Migrations Table Schema Fix	03 Jun 2026
v1.5.34	Dynamic Port in Server Setup	03 Jun 2026
v1.5.35	Music Pause During Announcements	03 Jun 2026
v1.5.36	Version Display in Admin Settings	13 Jul 2026
v1.5.45	Audio Fixes & Standalone Screens	04 Jul 2026
v1.5.46	Archival, Backups & Licensing	04 Jul 2026
v1.5.47	Branch Merge (v1.5.36 + v1.5.46)	12 Oct 2026
v1.5.48	Timezone & Ticket Numbering Fixes	12 Oct 2026
v1.5.49	New License Signing Key	12 Oct 2026
v1.5.50	Staff Setup Test Connection Fix	13 Oct 2026
v1.6.0	LAN Server Discovery & Port Conflicts	13 Oct 2026
v1.7.0	Offline Recorded Voice Announcements	13 Oct 2026
v1.8.0	Voice Engine & Spoken Announcements	14 Oct 2026
v1.9.0	Choose the Announcement Voice	14 Oct 2026

Detailed Release Notes

Every change, fix, and improvement — file level detail for all versions

v1.2.0 INITIAL RELEASE

06 May 2026

commit 45291daa

Initial Release — Full queue management system for Al-Noor International School

Files changed: 73 · +16712 " 0lines

```
Initial commit — SchoolQ v1.2.0

Queue management system for Al-Noor International School.

- Electron app (server + staff installers)
- Node/Express backend with SQLite
- React/Vite frontend
- Bilingual TTS (Azure Neural / Google Translate fallback)
- Arabic + English announcement support
- Custom voice templates in settings
```

APPLICATION ARCHITECTURE

- Electron 41 desktop application for Windows (x64)
  - Server build: includes backend + frontend + SQLite database
  - Staff build: frontend only — connects to server over LAN
- Node.js / Express backend running on port 3000
- SQLite database via better-sqlite3 (WAL mode, foreign keys enabled)
- React 19 + Vite 8 frontend with Tailwind CSS v3
- Socket.io v4 for real-time bidirectional communication
- JWT-based authentication with bcrypt password hashing

DATABASE SCHEMA (INITIAL TABLES)

- users — user_id, username, password_hash, full_name, role, department_id, is_active, created_at, last_login
- departments — department_id, name, code, color_code, display_order, is_active, created_at
- service_categories — category_id, department_id, name, estimated_time_minutes, is_active
- tickets — ticket_id, ticket_number, department_id, category_id, parent_name, student_name, student_id, phone, purpose, priority, status, created_at, called_at, completed_at, service_duration, served_by_user_id, notes, call_count
- settings — setting_key (PK), setting_value, setting_type
- announcements — announcement_id, message_text, message_text_ar, is_active, display_order
- audit_logs — log_id, user_id, action, entity_type, entity_id, details, logged_at
- Indexes: idx_tickets_dept_status, idx_tickets_created, idx_tickets_number

ROLE SYSTEM

- super_admin — full system access, can manage all settings, users, departments
- admin — queue management, reports, limited settings
- staff — queue dashboard for their assigned department only
- reception — ticket creation only

QUEUE FEATURES

- Ticket creation at reception: parent name, student name, student ID, phone, purpose, priority
- Priority levels: Regular, Urgent, Elderly (Priority), VIP — shown with color coding and flags on ticket number
- Ticket number format: DEPT-YYYYMMDD-NNN[flag] (e.g. REG-20260506-001U for urgent)
- Staff queue dashboard: call next, recall, complete, skip (with reason), no-show, transfer to department
- Public display screen: live current ticket, department queue status, announcement ticker
- Ticket receipt printed/displayed on creation with QR-ready data

VOICE ANNOUNCEMENT SYSTEM

- TTS endpoint: GET /api/tts?text=...&lang=en|ar
- Primary engine: Azure Neural TTS (requires AZURE_SPEECH_KEY and AZURE_SPEECH_REGION in .env)
- Fallback engine: Google Translate TTS (mp3 stream, no API key required)
- Template variables: {ticket} = ticket number, {department} = department name
- Language modes per announcement: English only, Arabic only, or Both
- Announcements scroll in a ticker on the public display screen

ADMINISTRATION PANEL

- Departments: create, edit, deactivate — with color picker, display order, service categories
- Users: create, edit, deactivate, assign to department, set role
- Announcements: create, edit, toggle active, set language, display order
- Settings: school name, primary color, working hours, TTS templates, no-show threshold, Azure keys
- Reports: daily summary — tickets served, waiting, no-shows, average service time per department

ELECTRON / BUILD

-

Two electron-builder configs: config/eb-server.json and config/eb-staff.json

- NSIS installer (non-oneClick, allow directory choice, desktop + start menu shortcuts)
- preload.js exposes ipcRenderer.invoke/send for config, IP detection, server ping
- Server build reads DB_PATH from Electron userData to avoid writing to Program Files

v1.3.0 PATCH

06 May 2026
commit 0eb86f53

Bilingual TTS Fix & Voice Quality Improvements

Files changed: 2 · +82 " 45lines

```
Fix bilingual TTS and improve voice robustness

- PublicDisplay: replace isSpeaking boolean flag with AbortController pattern
  — each speak() call aborts previous audio + Web Speech, no concurrent drain loops
  — adds 700ms pause between EN and AR for natural bilingual delivery
  — playTTS now has 12s hard timeout, uses {once:true} listeners, and auto-picks
    best neural voice (Microsoft Online/Natural) from Web Speech fallback
  — preloads browser voices on mount so first announcement uses best voice
- tts.js: validate Google TTS response is real MP3 (sync byte check); throws
  instead of silently serving HTML when Google blocks the request
```

FILES CHANGED

File	+Lines	" Lines	Note
src/frontend/src/pages/PublicDisplay.jsx	+116	" 43	AbortController pattern, 700ms pause, voice preload
src/backend/routes/tts.js	+7	" 7	MP3 byte validation
config/eb-server.json	+2	" 1	Version bump 1.2.0!'1.3.0
config/eb-staff.json	+2	" 1	Version bump 1.2.0!'1.3.0

TECHNICAL DETAILS — PUBLICDISPLAY.JSX

- Problem: isSpeaking boolean flag did not prevent overlapping audio when two tickets were called quickly
- Root cause: audio.play() resolves immediately on playback START, not on END — previous approach treated play() as a blocking call
- Fix: AbortController — each new speak() call cancels the previous audio element and Web Speech utterance before starting a new one
- Added 700ms silence gap between English and Arabic for natural bilingual delivery
- playTTS() now has a 12-second hard timeout via AbortController signal — prevents stuck audio blocking the queue
- Added {once: true} to all event listeners — prevents listener accumulation across repeated calls
- Voices preloaded on mount: window.speechSynthesis.getVoices() called at component startup
  - Selects Microsoft Online Natural voice if available (highest quality Web Speech)
  - Falls back to any en-US or ar-SA voice, then the browser default

TECHNICAL DETAILS — TTS.JS

- Problem: Google occasionally returns an HTML error page instead of MP3 when rate-limited
- Fix: reads first 3 bytes of response synchronously — valid MP3 starts with ID3 or 0xFF 0xFB
- If response is HTML (starts with <), throws 502 Bad Gateway instead of serving garbled audio
- Frontend correctly falls back to Web Speech Synthesis on 502

v1.3.1 SECURITY

07 May 2026
commit 61c35f42

Security Hardening, Performance Improvements & Code Cleanup

Files changed: 13 · +223 " 487lines

```
Release v1.3.1 — security hardening, performance improvements, cleanup

- CORS: restrict to localhost + private LAN IPs (was open wildcard)
- Auth: add rate limiting to login (20 req / 15 min)
- Tickets: wrap number generation in a transaction to prevent duplicates
- Settings: add in-memory cache (30s TTL) to avoid per-request DB queries
- Server: add gzip compression and morgan request logging
- DB: add morgan, compression, express-rate-limit dependencies
- School name fallback now consistent across queueController and display route
- Deleted unused legacy pages: AdminPanel.jsx, DisplayMonitor.jsx
- README: add security checklist for production deployment
```

FILES CHANGED

File	+Lines	" Lines	Note
src/backend/server.js	+30	" 1	CORS, rate limit, compression, morgan
src/backend/routes/auth.js	+11		loginLimiter middleware applied
src/backend/controllers/ticketController.js	+22		db.transaction() wrapping
src/backend/database/settingsCache.js	+23		& NEW FILE
src/backend/controllers/queueController.js	+7	" 1	Use settingsCache for school_name
src/backend/routes/display.js	+6	" 1	Use settingsCache for school_name
src/backend/routes/admin.js	+2		settingsCache invalidation on settings save
package.json	+9	" 1	Added: morgan, compression, express-rate-limit
src/frontend/src/pages/AdminPanel.jsx		" 292	DELETED — legacy unused page
src/frontend/src/pages/DisplayMonitor.jsx		" 166	DELETED — legacy unused page
README.md	+16	" 1	Security deployment checklist

SECURITY — CORS

- Before: CORS allowed any origin (wildcard *) — any website on any network could call the API
- After: regex restricts to localhost, 127.0.0.1, and private LAN ranges only:
  - 192.168.x.x (home/school LAN)
  - 10.x.x.x (corporate LAN)
  - 172.16.x.x – 172.31.x.x (Docker / VPN ranges)
- No-origin requests (same-origin, Electron) still allowed

SECURITY — RATE LIMITING

- express-rate-limit applied to POST /api/auth/login endpoint
- Limit: 20 requests per 15-minute window per IP address
- Response on limit: HTTP 429 — "Too many login attempts. Try again in 15 minutes."
- Standard rate-limit response headers added (X-RateLimit-*)

PERFORMANCE — SETTINGS CACHE

- settingsCache.js: Map-based in-memory cache with 30-second TTL
- Used by queueController (school_name for announcements) and display route
- Reduces DB queries from every-request to once per 30 seconds for high-read settings
- cache.get(key) / cache.set(key, value) / cache.invalidate() API
- Admin settings save calls cache.invalidate() to force fresh read on next request

PERFORMANCE — COMPRESSION & LOGGING

- compression middleware added — gzip/brotli encoding on all HTTP responses
- Reduces response size by ~65–80% for JSON payloads, ~75% for frontend JS bundles
- morgan "dev" format added — logs HTTP method, path, status, response time to console

RELIABILITY — DUPLICATE TICKET PREVENTION

- Before: ticket number generation and INSERT were two separate DB operations — race condition possible under concurrent requests
- After: wrapped in db.transaction() — atomic read-generate-insert prevents duplicate ticket numbers
- Transaction runs in IMMEDIATE mode (SQLite default) — serialized write access

v1.3.x BUG FIX

07 May 2026
commit multiple

Bug Fix Series — 5 Patches (Token, No-Show Setting, Startup, ABI, Final Call)

Files changed: 14 · +79 " 40lines

FIX 1 — JWT TOKEN EXPIRY & AUTO-LOGOUT ON 401 [BDC02E5E]

File	+Lines	" Lines	Note
src/backend/controllers/authController.js	+2	" 1	expiresIn: 8h !' 24h
src/frontend/src/lib/api.js	+12		Response interceptor for 401 handling

- Problem: Staff logging in at 7:00 AM were being logged out mid-day due to 8h token expiry
- Fix: JWT expiry extended from 8h to 24h — covers a full school day regardless of login time
- Problem: Expired/invalid tokens showed cryptic browser error "Invalid token" with no recovery path
- Fix: axios response interceptor in api.js — any 401 response triggers:
  - localStorage.removeItem("token")
  - localStorage.removeItem("user")
  - window.location.href = "/login" — automatic redirect

FIX 2 — NO-SHOW AFTER X CALLS WAS HARDCODED TO 3 [2C060704]

File	+Lines	" Lines	Note
src/backend/controllers/queueController.js	+7	" 2	Reads from settings DB instead of const 3
src/backend/server.js	+1		Expose no_show_after_calls in public settings
src/frontend/src/pages/QueueDashboard.jsx	+14	" 4	Dynamic noShowAfterCalls state

- Problem: noShow() controller had hardcoded value 3 — the admin-configurable setting was being ignored
- Fix: queueController reads no_show_after_calls from settings table on every call
- Error message now shows actual value: "Ticket has already received 5 calls (limit: 5)"
- QueueDashboard fetches no_show_after_calls from /api/settings/public on mount
- No-show button disabled state and tooltip text use the fetched value instead of hardcoded 3
- Public settings endpoint updated to include no_show_after_calls key

FIX 3 — ELECTRON STARTUP SILENT CRASH [BBE52776]

File	+Lines	" Lines	Note
electron/main.js	+44	" 31	Error dialog, waitForServer timeout 20s!'30s

- Problem: any require() failure (missing module, DB error, port conflict) caused a completely silent crash
- Symptom: Electron process was alive in Task Manager but no window ever appeared
- Fix: startLocalServer() wrapped in try/catch — all errors show a dialog.showErrorBox() before app.quit()
- Error dialog shows: error type + message + "Please reinstall or contact Vyra Systems"
- waitForServer timeout increased from 20s to 30s for slower school machines

FIX 4 — NATIVE MODULE ABI MISMATCH [7900EC52]

File	+Lines	" Lines	Note
config/eb-server.json	+1		buildDependenciesFromSource: true
config/eb-staff.json	+1		buildDependenciesFromSource: true

- Problem: better-sqlite3 and bcrypt pre-built for system Node.js (ABI 137) — crashed in Electron 41 (ABI 145)
- Error: "The module was compiled against a different Node.js version"
- Fix: buildDependenciesFromSource: true in both builder configs
- @electron/rebuild recompiles bcrypt and better-sqlite3 from C++ source for the exact Electron ABI at build time
- Build time increases by ~2 minutes but produces correct binaries — no more crash on launch

FIX 5 — FINAL CALL ALWAYS USED FINAL TEMPLATE [5A49BCB2]

File	+Lines	" Lines	Note
src/backend/controllers/queueController.js	+4	" 1	is_final logic added

src/frontend/src/pages/PublicDisplay.jsx	+2	" 1	Reads is_final from socket data

- Problem: every ticket recall used the "final call" TTS template, not just the last one
- Fix: backend computes is_final = call_count >= no_show_after_calls after incrementing call_count
- Socket emit for ticket_recalled now includes is_final boolean
- PublicDisplay.jsx uses is_final to choose between recall_template_en/ar and final_call_template_en/ar

v1.4.0 MINOR UPDATE

07 May 2026
commit 8cedfca0

Per-User Screen Permissions, School Logo Upload & Full Audit Log

Files changed: 12 · +125 " 27lines

```
WIP v1.4.0 — permissions, logo, audit log, menu bar removed

- Remove Electron menu bar (File/Edit/View)
- Logo upload in admin settings, stored as base64, shown on public display
- Per-user screen access control (allowed_pages JSON column on users)
- Admin Users form now has Screen Access checkboxes
- ProtectedRoute enforces allowed_pages when set
- Audit log helper wired into auth + ticket creation
- DB migration: adds allowed_pages to users table
- JSON body limit raised to 5mb for base64 logo uploads
```

FILES CHANGED

File	+Lines	" Lines	Note
src/backend/audit.js	+13		& NEW FILE
src/frontend/src/pages/admin/Users.jsx	+38	" 2	Screen Access checkboxes UI
src/frontend/src/pages/admin/Settings.jsx	+29	" 1	Logo upload + preview UI
src/frontend/src/App.jsx	+19	" 6	ProtectedRoute allowed_pages enforcement
src/backend/controllers/authController.js	+13	" 3	allowed_pages in JWT response, audit login
src/backend/routes/admin.js	+14	" 5	allowed_pages CRUD, logo setting
src/frontend/src/pages/PublicDisplay.jsx	+15	" 4	School logo display
src/backend/database/db.js	+1		Migration: allowed_pages column
src/backend/server.js	+4	" 2	JSON body limit 1mb!'5mb
src/backend/controllers/queueController.js	+1		Audit ticket_called event
src/backend/controllers/ticketController.js	+2		Audit TICKET_CREATED event
electron/main.js	+3	" 1	Menu.setApplicationMenu(null)

SCREEN ACCESS CONTROL

- New column: users.allowed_pages (TEXT, JSON array, DEFAULT NULL)
- NULL means no restriction — all screens accessible (backwards compatible with existing users)
- Example value: ["reception","queue","reports"]
- ProtectedRoute logic: if page prop is set AND user.allowed_pages is not null, check inclusion
  - If page is in allowed_pages !' grant access
  - If page not in allowed_pages !' redirect to /home
  - super_admin and admin: allowed_pages is always null !' full access
- Admin Users form: checkboxes for Reception, Queue, Admin, Reports screens
- PUT /admin/users/:id accepts allowed_pages array, JSON.stringify() before storing
- GET /api/auth/me includes allowed_pages in response for client-side routing decisions

SCHOOL LOGO

- New setting key: school_logo (TEXT, stores data: base64 URI)
- Admin Settings: file input (accept image/*, max 2MB), FileReader converts to base64
- JSON body limit raised from 1MB to 5MB in express.json() to handle base64-encoded images
- Public display screen shows logo in top-left corner of the display
- Logo included in public settings endpoint for frontend access without auth

AUDIT LOG SYSTEM

- audit.js: singleton prepared INSERT statement — INSERT INTO audit_logs (user_id, action, entity_type, entity_id, details)
- log(userId, action, entityType, entityId, details) — details JSON.stringify()'d
- Audited actions: LOGIN, TICKET_CREATED, TICKET_CALLED, TICKET_COMPLETED, NO_SHOW, TRANSFER, USER_CREATED, USER_UPDATED, DEPT_CREATED, DEPT_UPDATED, ANNOUNCEMENT_CREATED, ANNOUNCEMENT_DELETED, SETTINGS_UPDATED
- Audit failures silently swallowed (catch {}) — audit log write errors do not break primary operations
- Admin panel Audit Log page: date range filter, user filter, action filter, paginated table, export

ELECTRON

- Menu.setApplicationMenu(null) — removes File/Edit/View/Window/Help menu bar from all windows
- Provides a cleaner kiosk-style appearance appropriate for school deployment

v1.5.15 MINOR UPDATE

15 May 2026
commit 384d5b4a

Service Types Deep Integration, Room Numbers & Sequential Bilingual Announcements

Files changed: 27 · +1526 " 422lines

```
v1.5.15 — service types deep integration, room number, announcement fixes

- Part 1: Reception live wait estimate using per-service estimated_time_minutes
- Part 2: Service Type detailed report tab with estimated vs actual chart + export
- Part 3: Smart estimate suggestions in Departments modal from last 30 days real data
- Add room_number field to departments, {room} variable in voice templates
- Fix announcement TTS: both languages now play sequentially (English then Arabic)
- Fix announcement ticker to use per-announcement speak_language field
- Fix Test Connection in Staff setup via IPC (bypasses CORS from file://)
- Fix screen access permissions, toggleActive wipe bug, Home tile filter order
- Fix port conflict detection with isSchoolQOnPort3000 HTTP check
```

FILES CHANGED

File	+Lines	" Lines	Note
src/frontend/src/pages/Reception.jsx	+194	" 134	Complete rewrite — live wait estimate
src/frontend/src/pages/Reports.jsx	+459	" 284	Complete rewrite — tab system, service types
src/frontend/src/pages/admin/Departments.jsx	+307	" 82	Smart suggestions, room number form
src/frontend/src/pages/admin/QueueControl.jsx	+311		& NEW FILE
src/frontend/src/pages/admin/AuditLog.jsx	+139		& NEW FILE
src/backend/routes/reports.js	+64		& NEW FILE
src/frontend/src/App.jsx	+177	" 58	Route additions, ProtectedRoute fixes
src/backend/routes/admin.js	+81	" 1	room_number CRUD, category CRUD
src/frontend/src/pages/admin/Announcements.jsx	+53	" 10	Sequential bilingual playback fix
src/frontend/src/pages/PublicDisplay.jsx	+29	" 9	applyVars {room}, per-announcement language
src/backend/controllers/queueController.js	+10		department_room in socket emits
src/frontend/src/lib/api.js	+13		reportsAPI export added
src/backend/database/db.js	+2		room_number migration
electron/main.js	+42	" 1	IPC test-server handler, port check fix
src/frontend/src/store/useAuthStore.js	+5		updateUser action
src/frontend/src/pages/admin/Users.jsx	+15	" 2	Toggle fix
src/frontend/src/pages/admin/Settings.jsx	+3	" 1	{room} hint added
electron/preload.js	+3		testServer IPC bridge

PART 1 — LIVE WAIT ESTIMATE AT RECEPTION

- Two new useEffect hooks: one loads service categories + current queue count when department changes; one recalculates estimate when category or count changes
- Formula: waitMinutes = Math.ceil(waitingCount × estimatedTimeMinutes / activeStaffCount)
- Uses selected service category's estimated_time_minutes (not a hardcoded value)
- Falls back to department average estimated time if no category selected
- Service type dropdown shows "(X min)" label beside each service name
- Wait estimate banner examples:
  - "~15 min — 3 people ahead × 5 min/service"
  - "You're next — no wait!" — shown when queue is empty
  - "~2 min — 1 person ahead" — for short queues

PART 2 — SERVICE TYPE REPORT TAB

- Reports.jsx rewritten with a tab system: "Daily Summary" | "Service Types"
- New DailySummary and ServiceTypeReport sub-components
- Service Types tab — filters: date range (from/to), department (optional)
- Table columns: Service Type, Department, Estimated (min), Actual Avg (min), Accuracy, Total, Served, No-Shows
- AccuracyBadge: green "d2 min diff, yellow "d5 min, red >5 min
- Recharts grouped bar chart: amber bars = Estimated, teal bars = Actual, per service type
- Export to PDF via jsPDF + jspdf-autotable; Export to Excel via xlsx library
- Backend: GET /reports/service-types?from=&to=&department_id= — joins tickets with service_categories
- Backend: GET /reports/category-stats — actual_avg_minutes and sample_count per category (last 30 days)

PART 3 — SMART ESTIMATE SUGGESTIONS

- Departments modal fetches reportsAPI.getCategoryStats() when opened
- Builds catStats Map: category_id !' { actual_avg_minutes, sample_count }
- When editing a service type row, the catStats for that category are shown:
  - diff "d 2 min: green "' your estimate is accurate" — no action needed
  - diff > 2 min: amber banner "Actual avg last 30 days: X min (N tickets)" + "Use X min" button
- "Use X min" button calls setcf("estimated_time_minutes", actualAvg) — updates the form field immediately
- Sample count shown so admin knows whether data is statistically reliable

ROOM NUMBER FEATURE

- DB migration: ALTER TABLE departments ADD COLUMN room_number TEXT DEFAULT NULL
- Admin Departments form: new "Room Number" text input with hint: "Use {room} in voice templates"
- PUT /departments/:id and POST /departments now accept and save room_number
- GET /departments/:id returns room_number in response
- queueController: ticket_called and ticket_recalled socket emits include department_room: dept.room_number || null
- PublicDisplay.jsx applyVars() updated: replace({room}) replaces with departmentRoom || empty string
- Settings page TTS template hints updated: Variables: {ticket}, {department}, {room}

ANNOUNCEMENT SEQUENTIAL PLAYBACK FIX

- Problem: admin Test Voice button for "both" language only played English
- Root cause: handleTest used ternary (lang === "ar" ? ar : en) — "both" fell through to English only
- Fix: explicit if/else if/else — both case: await playTTS(en), then await playTTS(ar)
- Problem: both languages were playing simultaneously (overlapping audio)
- Root cause: audio.play() resolves when playback STARTS (not ends) — awaiting it does not wait for finish
- Fix: playTTS() returns a Promise that resolves on the "ended" event:
  - audio.addEventListener("ended", resolve, { once: true })
  - audio.addEventListener("error", resolve, { once: true }) — also resolves on failure to avoid hanging
  - audio.play().catch(() => audio.dispatchEvent(new Event("error"))) — surfaces play() rejection

OTHER BUG FIXES (V1.5.15)

- Staff setup Test Connection: was failing with CORS error because fetch() from file:// has null origin
  - Fix: IPC bridge — preload.js exposes testServer(), electron main.js handles via http.get to avoid CORS
- toggleActive bug: saving user with the toggle button was wiping allowed_pages (spread order issue)
- Home page tile ordering fixed — admin/super_admin see different tile sets in correct order
- Port conflict detection: previously any TCP listener on 3000 was killed
  - Fix: isSchoolQOnPort3000() HTTP check — only kill if the process on 3000 is NOT SchoolQ

v1.5.16 SECURITY

20 May 2026

commit not committed (working changes)

Force Logout on Deactivation, Socket Authentication & Auth Middleware Hardening

Files changed: 5 · +120 " 18lines

FILES CHANGED

File	+Lines	" Lines	Note
src/backend/socket/handlers.js	+55	" 17	JWT auth middleware, user-socket tracking,
src/backend/middleware/auth.js	+5	" 2	is_active DB check on every request
src/backend/routes/admin.js	+8		forceLogout() call on user deactivation
src/frontend/src/lib/useSocket.js	+18	" 4	token in auth, force_logout listener
src/backend/controllers/authController.js	+2	" 1	JWT expiry 24h!'8h

FORCE LOGOUT FLOW

- Admin deactivates user !' PUT /admin/users/:id sets is_active = 0 !' socketHandlers.forceLogout(userId) called
- handlers.js maintains userSockets Map: userId !' Set<socketId> — multiple tabs tracked per user
- forceLogout(io, userId): iterates the Set, calls io.to(socketId).emit("force_logout") for each
- Frontend useSocket hook listens: socket.on("force_logout", forceLogoutCleanup)
- forceLogoutCleanup(): localStorage.removeItem("token"), removeItem("user"), window.location.href = "/login"
- Result: deactivated user's browser redirects to /login within ~100ms of admin clicking save

SOCKET AUTHENTICATION

- io.use() middleware added in handlers.js — runs before any socket is accepted
- Reads token from socket.handshake.auth.token
  - If no token !' next() with no user set (allows public display to connect unauthenticated)
  - If token present !' jwt.verify() + DB is_active check !' socket.userId = decoded.user_id
  - If token invalid or user inactive !' next(new Error("...")) — connection rejected
- useSocket.js passes token in connection options: { auth: { token: localStorage.getItem("token") } }
- useMonitorSocket (public display) passes no auth — intentionally unauthenticated
- Socket tracking: trackSocket(userId, socketId) on connect, untrackSocket on disconnect

AUTH MIDDLEWARE — ACTIVE CHECK

- Before: auth.js only verified JWT signature — deactivated users could make API calls until token expired
- After: additional DB query after token decode: SELECT is_active FROM users WHERE user_id = ?
- If is_active = 0 or user not found !' 401 "Account deactivated"
- api.js 401 interceptor catches this and redirects to /login
- Performance note: indexed on users(is_active) — fast lookup, minimal overhead per request

JWT EXPIRY

- Reduced from 24h back to 8h — 24h was a temporary fix for the early-login problem
- Now that force-logout and is_active check exist, short expiry is safe and preferable
- 8h matches a standard school work day; session always ends at logout or token expiry

v1.5.17 SECURITY

20 May 2026
commit not committed (working changes)

Phases 1–4: Input Validation, Per-Install JWT Secret, Rate Limiting, Error Handler & DB Indexes

Files changed: 13 · +280 " 25ines

FILES CHANGED

File	+Lines	" Lines	Note
src/backend/middleware/validate.js	+65		& NEW FILE
src/backend/middleware/errorHandler.js	+11		& NEW FILE
src/backend/routes/auth.js	+6	" 2	Validation on login, change-password
src/backend/routes/tickets.js	+4	" 1	Validation on POST /tickets
src/backend/routes/admin.js	+12	" 6	Validation on all POST/PUT routes
src/backend/server.js	+16	" 2	mutationLimiter, errorHandler registration
src/backend/database/schema.sql	+9	" 3	7 new CREATE INDEX statements
src/backend/controllers/authController.js	+1	" 1	Password min 6!'8 chars
src/backend/audit.js	+2	" 1	Silent catch {} !' logged error
electron/main.js	+12	" 2	Per-install JWT secret generation
config/eb-server.json		" 1	Removed .env from build files list
package.json	+2	" 1	express-validator dependency
src/backend/database/schema.sql	+9	" 3	7 new indexes

PHASE 1 — INPUT VALIDATION (EXPRESS-VALIDATOR)

- validate.js: validation chains using body(), param(), bail() from express-validator
- checkValidation middleware: calls validationResult(req) — returns HTTP 422 with first error message on failure
- All validation chains use .bail() — stops on first error per field, avoids cascading messages
- Rules defined:
  - rules.login — username: trim + notEmpty; password: notEmpty
  - rules.changePassword — current_password: notEmpty; new_password: isLength({ min: 8 })
  - rules.createUser — username: 3–50 chars; password: "e8 chars; full_name: notEmpty; role: isIn([valid_roles])
  - rules.updateUser — full_name: notEmpty; role: isIn([valid_roles])
  - rules.createDepartment — name: notEmpty; code: isAlphanumeric + isLength(1–10)
  - rules.updateDepartment — same as create
  - rules.createAnnouncement — message_text: notEmpty; speak_language: optional isIn(["en","ar","both"])
  - rules.createTicket — department_id: isInt({ min:1 }); priority: optional isIn(VALID_PRIORITY); names/phone: optional length limits
- Applied to: POST /auth/login, PUT /auth/change-password, POST+PUT /admin/users, POST+PUT /admin/departments, POST+PUT /admin/announcements, POST /tickets

PHASE 2 — SECURITY HARDENING

- Per-install JWT secret — electron/main.js startLocalServer():
  - Checks for jwt-secret.txt in app.getPath("userData")
  - If absent: crypto.randomBytes(64).toString("hex") !' writes to jwt-secret.txt
  - Sets process.env.JWT_SECRET from file before requiring backend server
  - Result: each installation has a cryptographically unique 512-bit secret
  - Removed dotenv call — JWT_SECRET no longer comes from .env
- .env removed from electron-builder files list in eb-server.json and root package.json build config
  - Before: JWT_SECRET was hardcoded "school-queue-secret-2024" and bundled into every installer
  - After: .env not included in build; secret generated fresh on first run
- mutationLimiter added in server.js: 60 requests / 1 minute / IP
  - Applied to: /api/tickets, /api/queue, /api/admin (all write-heavy endpoints)
  - Login retains its own stricter limiter (20 / 15 min)
- Minimum password: 6 !' 8 characters in authController.changePassword validation

PHASE 3 — CENTRALIZED ERROR HANDLER

- errorHandler.js: standard Express error-handling middleware signature (err, req, res, next)
- status < 500: pass err.message to client (intentional 4xx errors)
- status >= 500: log full error to console with method + path; return generic "Internal server error" to client
- Never exposes stack traces to clients — production-safe
- Registered as last middleware in server.js: app.use(errorHandler)
-

audit.js: catch {} !’ catch (err) { console.error("[AUDIT] Failed to write audit log:", action, err.message) }
  - Audit failures now visible in server logs but still do not break primary operations

PHASE 4 — DATABASE INDEXES

- 7 new indexes added to schema.sql with CREATE INDEX IF NOT EXISTS — safe to apply to existing databases:
  - idx_tickets_completed_at ON tickets(completed_at) — date range queries in service type reports
  - idx_tickets_served_by ON tickets(served_by_user_id) — per-staff performance queries
  - idx_tickets_category ON tickets(category_id) — service type report JOIN
  - idx_audit_logged_at ON audit_logs(logged_at) — audit log date filtering
  - idx_audit_user ON audit_logs(user_id) — per-user audit trail
  - idx_users_dept ON users(department_id) — staff count per department
  - idx_users_active ON users(is_active) — auth middleware active-check lookup
- Pre-existing indexes retained: idx_tickets_dept_status, idx_tickets_created, idx_tickets_number
- Impact: report queries covering 30+ day date ranges run significantly faster on databases with 1000+ tickets

v1.5.18 UX IMPROVEMENT

20 May 2026
commit not committed (working changes)

Phases 5–7: Toast Notifications, Skeleton Loaders, Migration Runner & Automated Tests

Files changed: 17 · +420 " 45lines

FILES CHANGED

File	+Lines	" Lines	Note
src/frontend/src/store/useToastStore.js	+22		& NEW FILE
src/frontend/src/components/Toast.jsx	+34		& NEW FILE
src/backend/middleware/validate.js			Already in v1.5.17
src/frontend/src/App.jsx	+3		<Toast /> mounted + import
src/frontend/tailwind.config.js	+5	" 2	fade-in keyframe + animation
src/frontend/src/pages/QueueDashboard.jsx	+22	" 7	Toast, skeleton, loading state
src/frontend/src/pages/Reception.jsx	+3	" 2	alert !' toast
src/frontend/src/pages/Reports.jsx	+3	" 2	alert !' toast (2 occurrences)
src/frontend/src/pages/admin/QueueControl.jsx	+8	" 7	alert !' toast (7 occurrences)
src/frontend/src/pages/admin/Users.jsx	+2	" 1	alert !' toast
src/frontend/src/pages/admin/Settings.jsx	+3	" 2	alert !' toast (2 occurrences)
src/backend/database/migrations/runner.js	+55		& NEW FILE
src/backend/database/db.js	+3	" 6	Use runner.js, remove try/catch blocks
src/backend/tests/validate.test.js	+45		& NEW FILE
src/backend/tests/auth.test.js	+72		& NEW FILE
src/backend/tests/tickets.test.js	+58		& NEW FILE
package.json	+8	" 1	jest, supertest, pdfkit, npm test script

PHASE 5 — TOAST NOTIFICATION SYSTEM

- useToastStore.js (Zustand): toasts array, addToast(message, type, duration), removeToast(id)
  - Auto-dismiss: setTimeout removes toast after duration (4000ms for errors, 3000ms for success/info)
  - Monotonic ID counter — no duplicate keys even under rapid toasts
- Toast.jsx: fixed bottom-right overlay, z-index 9999, stacks vertically with 2px gap
  - Types: error (red bg), success (green bg), info (blue bg)
  - Icons: ' for error, ' for success, !9 for info
  - Dismiss button (×) on each toast — calls removeToast(id)
  - animate-fade-in CSS class: opacity 0!'1 + translateY 8px!'0, duration 0.2s ease-out
- toast helper object: toast.error(msg), toast.success(msg), toast.info(msg)
  - Uses useToastStore.getState().addToast() — callable outside React components (from catch blocks)
- <Toast /> mounted in App.jsx root — single instance, always visible regardless of current route
- 14 alert() calls replaced across 6 files — all errors now non-blocking, dismissable, auto-clearing

PHASE 5 — LOADING SKELETON IN QUEUEDASHBOARD

- QueueSkeleton component: 3 animated gray bars (h-16 bg-gray-200 animate-pulse rounded-lg)
- loading state (useState(true)) — set to false in fetchQueue finally block
- Queue table hidden (CSS class "hidden") while loading = true; skeleton shown instead
- Prevents flash of "No waiting tickets" empty state on initial load
- animate-pulse is a Tailwind built-in: opacity oscillates via CSS @keyframes

PHASE 6 — PROPER MIGRATION RUNNER

- migrations/runner.js replaces fragile try/catch ALTER TABLE pattern in db.js
- migrations table: CREATE TABLE IF NOT EXISTS migrations (id INTEGER PRIMARY KEY, name TEXT, applied_at TIMESTAMP)
- On startup: loads applied Set from migrations table, iterates migrations array, skips applied ones
- Each migration: { id: number, name: string, up: (db) => void }
- Error handling: "duplicate column" error (SQLite-specific) !' marks as applied and continues
- Other errors !' logged to console, migration not marked applied (will retry next start)
- Console output per applied migration: ' Migration 1: add_announcement_speak_language
- 4 existing schema changes ported:
  - id:1 — add speak_language to announcements
  - id:2 — add allowed_pages to users
  - id:3 — add name_ar to departments
  -

id:4 — add room_number to departments
- Future changes: add { id: N, name: "description", up: (db) => db.prepare("...").run() } to migrations array

PHASE 7 — AUTOMATED TEST SUITE (16 TESTS)

- Jest + Supertest installed as devDependencies; npm test configured
- testMatch: **/tests/**/*.test.js — all test files in src/backend/tests/
- All tests mock the DB (jest.mock("../database/db")) — no Electron native module dependency
- Why mocking needed: better-sqlite3 is compiled for Electron's Node ABI — incompatible with test runner

validate.test.j
s 7 tests —
pure
middleware,
no DB needed

- createUser: username < 3 chars !' 422, invalid role !' 422, password < 8 chars !' 422, valid data !' 200
- createTicket: non-integer department_id !' 422, invalid priority !' 422, valid data !' 200

auth.test.js
5 tests —
auth routes
with
mocked DB

- login: missing username !' 422, missing password !' 422, wrong password !' 401, valid login !' 200 + token
- change-password: new_password < 8 chars !' 422

tickets.test.j
s 4 tests —
ticket routes
with mocked
DB

- no auth !' 401, missing department_id !' 422, invalid priority !' 422, valid data !' passes validation

- All 16 tests pass in ~2.7 seconds
- Test run command: npm test (runs npx jest)

v1.5.25 MINOR UPDATE

02 Jun 2026
commit dd236e0

Cancel / Left Action, No-Show Reason, Granular Report Permissions & Ticket Log Wait Time Fix

Files changed: 13 · +420 " 55lines

```
v1.5.25 — cancel action, no-show reason, granular report permissions, ticket log wait time fix

- Cancel/Left action: customer-left-voluntarily status with required reason
- No-Show now accepts optional reason stored in ticket notes
- Granular report tab permissions: restrict individual report tabs per user
- Ticket log wait_minutes shows elapsed time for currently-waiting tickets
```

CANCEL / LEFT ACTION

- New ticket status: cancelled — distinct from skip (back of queue) and no_show (did not answer calls)
- Staff clicks "CANCEL / LEFT" on serving panel or waiting row — required reason field shown
- Backend: PUT /queue/tickets/:id/cancel validates reason is not empty (422 if blank)
- Ticket updated: status = cancelled, completed_at = now(), notes = "Cancelled: <reason>"
- Available on both Queue Dashboard and All Queues screen
- Logged to audit log as TICKET_CANCELLED with reason included

NO-SHOW OPTIONAL REASON

- No-Show modal now includes an optional text field for reason
- When provided, stored in ticket notes as "No-Show: <reason>"
- Helps track patterns (e.g. parent left, wrong department, etc.)

GRANULAR REPORT TAB PERMISSIONS

- ALL_PAGES in Users.jsx expanded with 5 report sub-keys:
  - reports_daily, reports_service_types, reports_transfers, reports_purposes, reports_ticket_log
- Sub-items visually indented (ml-6) under the Reports parent in the Screen Access list
- togglePage auto-manages parent: checking any sub-key adds "reports"; removing last sub-key removes "reports"
- Reports.jsx filters visible tabs at runtime based on user.allowed_pages:
  - null !' all tabs (role defaults)
  - Has reports_* sub-keys !' only those tabs shown
  - Only "reports" parent key !' all tabs shown
- Active tab auto-corrects to first visible tab on mount if current tab is hidden
- "cancelled" status added to STATUS_STYLE (gray) and ticket log status filter dropdown

TICKET LOG WAIT TIME FIX

- Previously: wait_minutes only calculated after ticket was called (NULL for still-waiting tickets)
- Fix: SQL CASE expression — waiting tickets use julianday("now") - julianday(created_at) × 1440
- Live elapsed wait time now shown for currently waiting tickets in the Ticket Log report

File	+Lines	" Lines	Note
src/backend/controllers/queueController.js	+25	" 3	cancel action, noShow with reason
src/backend/routes/queue.js	+2		PUT /tickets/:id/cancel route
src/backend/routes/reports.js	+12	" 6	wait_minutes fix for waiting tickets
src/frontend/src/lib/api.js	+3	" 1	cancel(), noShow with reason param
src/frontend/src/pages/QueueDashboard.jsx	+68	" 18	cancel modal, no-show reason modal
src/frontend/src/pages/AllQueues.jsx	+40	" 12	cancel button + modal
src/frontend/src/pages/admin/Users.jsx	+18	" 4	report sub-permissions, indentation
src/frontend/src/pages/Reports.jsx	+22	" 8	tab filter, cancelled status

v1.5.26 UX IMPROVEMENT

03 Jun 2026
commit 33bc461

Danger Zone — Reset All Ticket Data Feature in Admin Settings

Files changed: 6 · +50 " 5lines

- "Danger Zone" section added to Admin Settings (visible to super_admin only)
- "Reset All Ticket Data" button with two-step confirmation: click !' "Are you sure?" !' "Yes, Delete All"
- Deletes all rows from tickets and transfers tables (users, departments, settings unaffected)
- Broadcasts queue_updated to all department Socket.io rooms immediately after reset
- Action logged in audit log as TICKETS_RESET

Backend: POST /admin/reset-tickets — requireRole super_admin
Frontend: Settings.jsx uses user.role check to gate Danger Zone section visibility

File	+Lines	" Lines	Note
src/backend/routes/admin.js	+16		POST /reset-tickets route
src/frontend/src/lib/api.js	+1		resetTickets() call
src/frontend/src/pages/admin/Settings.jsx	+43	" 1	Danger Zone UI, useAuthStore import

v1.5.27 MINOR UPDATE

03 Jun 2026
commit fc14665

Self-Service Kiosk Screen — Parents Get Tickets Without Reception Staff

Files changed: 10 · +310 " 3lines

```
v1.5.27 — self-service kiosk screen for parents

New /kiosk page: touch-optimized multi-step flow for parents to
get queue tickets without needing a reception staff member.
No login required — publicly accessible like the display screen.
```

KIOSK FLOW (4 STEPS)

- Welcome — "dV1bÖ(bvK · Welcome" full screen with large "Get a Ticket" button
- Info — Parent name (required) + student name (optional) — large touch-friendly inputs
- Department — Grid of large tappable department buttons
- Service Type — Grid of category buttons (only shown if dept has categories); "Other / Not sure" skip option
- Done — Ticket number displayed large, estimated wait shown, auto-resets after 30 seconds

TECHNICAL DETAILS

- /kiosk route added to App.jsx without ProtectedRoute (public, like /display)
- New backend route: POST /api/kiosk/ticket — no authentication required
  - Validates parent_name and department_id, always creates with priority = regular
  - Emits queue_updated to dept socket room — staff queues update instantly
  - Logs KIOSK_TICKET_CREATED to audit log with null user_id
- ticketController.js: exported createTicketTx and calculateWaitTime for reuse by kiosk route
- kioskAPI.createTicket() added to api.js frontend API layer
- Uses existing departmentAPI.getAll() and departmentAPI.getCategories() (already public endpoints)
- Settings (school name, logo) loaded from /api/settings/public on mount
- "Self-Service Kiosk" tile added to Home page for admin navigation (orange color)

File	+Lines	" Lines	Note
src/backend/routes/kiosk.js	+35		& NEW FILE
src/backend/server.js	+2		Register /api/kiosk route
src/backend/controllers/ticketController.js	+3		Export createTicketTx, calculateWaitTime
src/frontend/src/pages/Kiosk.jsx	+220		& NEW FILE
src/frontend/src/lib/api.js	+3		kioskAPI.createTicket()
src/frontend/src/App.jsx	+2		/kiosk route (public)
src/frontend/src/pages/Home.jsx	+7		Self-Service Kiosk tile

v1.5.28 BUG FIX

03 Jun 2026
commit 983a165

Auto Port Scan on Conflict — No Longer Kills Foreign Processes on Port 3000

Files changed: 6 · +54 " 37lines

- Old behavior: if port 3000 was busy, Electron forcibly killed the process holding it via PowerShell Stop-Process
- New behavior: server auto-scans ports 3000–3010, binds to first available, sets process.env.ACTUAL_PORT
- Electron main.js polls ACTUAL_PORT every 200ms until set, then opens window on that port
- Removed: isSchoolQOnPort3000(), killPort3000(), exec import from main.js
- server.js: server.on("error") catches EADDRINUSE — increments port and retries
- If all 10 ports busy: logs clear error and calls process.exit(1)
- Staff client URL config (http://server:3000) still works — staff always explicitly configure the port

File	+Lines	" Lines	Note
src/backend/server.js	+20	" 5	tryListen(), auto port scan, ACTUAL_PORT
electron/main.js	+18	" 32	waitForActualServer(), removed kill logic

v1.5.29 BUG FIX

03 Jun 2026
commit 7d746f6

Fix Temporal Dead Zone Crash — Cannot access createTicketTx before initialization

Files changed: 5 · +6 " 5ines

- Error: "Cannot access createTicketTx before initialization" on every startup
- Root cause: exports.createTicketTx = createTicketTx placed BEFORE the const createTicketTx = db.transaction(...) declaration
- JavaScript const is not hoisted — accessing it before declaration throws ReferenceError (TDZ)
- Fix: moved exports.createTicketTx = createTicketTx to AFTER the db.transaction() assignment
- One-line fix in ticketController.js — no functional change, just export order correction

v1.5.30 UX IMPROVEMENT

03 Jun 2026
commit 7b82dc4

Kiosk Auto-Print Ticket + Kiosk Permission in Screen Access

Files changed: 7 · +186 " 208lines

- Kiosk now auto-prints ticket immediately after parent selects service or department
- No manual "Print" button needed — print dialog opens automatically 600ms after ticket creation
- @media print CSS: kiosk screen hidden, TicketReceipt shown for printing
- @media screen: receipt div hidden on screen, only shown when printing
- afterprint event listener resets kiosk to Welcome screen (1.5s grace period)
- 45-second fallback timeout resets screen if printer is offline and afterprint never fires
- "kiosk" permission key added to ALL_PAGES in Users.jsx Screen Access section
- Home page kiosk tile uses page: "kiosk" — tile hidden when user has restricted access and kiosk not granted

File	+Lines	" Lines	Note
src/frontend/src/pages/Kiosk.jsx	+186	" 208	Auto-print, @media CSS, afterprint reset
src/frontend/src/pages/admin/Users.jsx	+1		kiosk key in ALL_PAGES
src/frontend/src/pages/Home.jsx	+1	" 1	page: null !' page: "kiosk"

v1.5.31 MINOR UPDATE

03 Jun 2026
commit bd09e57

Export/Import Users & Settings, Migrations Runner Restore, Scrollbar Fix

Files changed: 12 · +180 " 21lines

EXPORT/IMPORT USERS

- Admin Users page: "Export" button downloads JSON file of all users (with password_hash for migration)
- Admin Users page: "Import" button accepts JSON file — creates users that don't exist, skips existing usernames
- Import supports both password_hash (direct) and password (plain — auto-hashed via bcrypt)
- Results shown: "Imported X users, skipped Y"
- Endpoints: GET /admin/export/users, POST /admin/import/users — both require super_admin

EXPORT/IMPORT SETTINGS

- Admin Settings page: "Export Settings" downloads current settings as JSON
- "Import Settings" accepts a JSON file and applies all key-value pairs via existing saveSettings logic
- After import, local state updates to reflect new values without page reload

OTHER FIXES

- User form modal: added p-4 padding, max-h-[90vh], overflow-y-auto — form scrolls instead of getting cut off on small screens
- migrations/runner.js properly restored to source control (was gitignored by incorrect database/ pattern)
- .gitignore: database/ !' /database/ (only ignore root-level database/ folder, not src/backend/database/)

File	+Lines	" Lines	Note
src/backend/routes/admin.js	+36		Export/import users routes
src/frontend/src/lib/api.js	+2		exportUsers(), importUsers()
src/frontend/src/pages/admin/Users.jsx	+42	" 6	Export/Import UI buttons + handlers
src/frontend/src/pages/admin/Settings.jsx	+42	" 1	Export/Import settings UI + handlers
.gitignore	+1	" 1	/database/ fix
src/backend/database/migrations/runner.js	+44		& NEW FILE

v1.5.32 BUG FIX

03 Jun 2026
commit f168c7f

Fix Datatype Mismatch on Startup + Improved Error Dialog with Stack Trace

Files changed: 5 · +6 " 6ines

- Migration runner was using db.prepare(sql).run() for ALTER TABLE statements
- Correct API for DDL statements in better-sqlite3 is db.exec() — prepare().run() is for DML only
- Fix: changed to db.exec(migration.sql) in the for loop
- Error dialog in Electron now shows full stack trace (err.stack) instead of just err.message
- Stack trace appears in the "Failed to Start" dialog body — helps diagnose startup failures

v1.5.33 BUG FIX

03 Jun 2026
commit 404e94e

Fix Migrations Table INTEGER vs TEXT Schema Mismatch — Datatype Mismatch on Existing Databases

Files changed: 5 · +18 " 4ines

- Root cause: old database had migrations table created with id INTEGER PRIMARY KEY by an earlier version
- CREATE TABLE IF NOT EXISTS preserves the wrong schema — no error, table unchanged
- Then: INSERT INTO migrations (id) VALUES ("001_add_speak_language") fails — cannot insert TEXT into INTEGER PRIMARY KEY
- SQLite error: "datatype mismatch" — caught by Electron uncaughtException handler !' startup fails
- Fix: use PRAGMA table_info(migrations) to read the actual column type of "id"
- If type is not TEXT: DROP TABLE IF EXISTS migrations !' CREATE TABLE with correct TEXT PRIMARY KEY
- All migrations re-run (catching "duplicate column name" for already-applied ones) !' records inserted into new table
- Safe for both fresh installs and upgrades — no data loss (migrations table only tracks schema history)

File	+Lines	" Lines	Note
src/backend/database/migrations/runner.js	+14		PRAGMA type check, drop/recreate if wrong schema

v1.5.34 BUG FIX

03 Jun 2026
commit 5702888

Remove Hardcoded Port 3000 from Server Setup — Test Uses Real IP:Port

Files changed: 7 · +52 " 27ines

- ServerSetup.jsx: added Port input field (default 3000) alongside the server IP field
- "Test Connection" now hits ip:port, shows the exact address tested in the result message
- Save & Relaunch: server mode saves http://localhost:<window.location.port> (picks up auto-shifted port)
- Save & Relaunch: client mode saves http://<ip>:<entered-port>
- preload.js: testServer(ip, port) passes port through IPC channel
- main.js test-server handler: accepts port parameter, uses parseInt(port) || 3000
- On existing config load: correctly parses port from saved URL via new URL(cfg.serverUrl).port

File	+Lines	" Lines	Note
src/frontend/src/pages/ServerSetup.jsx	+25	" 18	Port field, dynamic URL building
electron/preload.js	+1	" 1	testServer(ip, port)
electron/main.js	+1	" 1	Port parameter in HTTP request

v1.5.35 UX IMPROVEMENT

03 Jun 2026
commit cffe6b0

Auto-Pause Background Music During Ticket Announcements via Media Key

Files changed: 7 · +49 " 8lines

- Problem: display PC plays background music in reception — TTS announcements mix with music
- Solution: send system-wide Media Play/Pause key before TTS, resume after
- Works with Spotify, YouTube, VLC, Windows Media Player — any player that responds to media keys
- On ticket call: music pauses !' TTS announcement plays !' music resumes automatically
- Rapid consecutive tickets handled correctly: musicPaused ref prevents double-pause
  - First ticket: pause music, play TTS, resume
  - Second ticket before first finishes: skip re-pause (already paused), play TTS, resume after last one
- media-key.ps1 written to userData on app startup — uses keybd_event(VK_MEDIA_PLAY_PAUSE = 0xB3)
- IPC: preload exposes mediaPlayPause(), main.js handles via execFile powershell
- No-op when display opened in a regular browser (window.electronAPI not available)

File	+Lines	" Lines	Note
electron/main.js	+18	" 1	media-key.ps1 write on ready, IPC handler
electron/preload.js	+1	" 1	mediaPlayPause() exposed
src/frontend/src/pages/PublicDisplay.jsx	+18	" 6	musicPaused ref, pre/post announce IPC call

v1.5.36 UX IMPROVEMENT

13 Jul 2026
commit 0a9d170

Version Number Displayed in Admin Settings Header

Files changed: 2 · +8 " 1lines

- vite.config.js: reads version from package.json at build time, injects as __APP_VERSION__ global constant
- Admin Settings page header now shows the running version as a navy/teal pill badge (e.g. "v1.5.36")
- Badge appears next to "System Settings" heading — always visible to super_admin
- No runtime API call needed — version is baked into the bundle at build time

File	+Lines	" Lines	Note
src/frontend/vite.config.js	+4		__APP_VERSION__ define
src/frontend/src/pages/admin/Settings.jsx	+6	" 1	Version badge in header

v1.5.45 MINOR UPDATE

04 Jul 2026
commit cccfe15

Announcement Audio Fixes, Single-Instance Server, Standalone Access-Controlled Screens

Files changed: 17 · +192 " 53lines

- Silent call/recall fixed — root cause: duplicate Server instances (ports 3000 + 3001) sharing one DB with isolated socket rooms
  - Server build now enforces a single instance; Staff/client builds may still open multiple windows
- Display plays announcements without a click (Electron autoplayPolicy)
- Music pause now works: Chromium hardware media-key capture disabled so Play/Pause reaches the external player
- On-screen audio/socket diagnostics removed from the public display
- Queue Control (/queue-control) and Announcements (/announcements) available outside the Admin panel
  - Shared ScreenShell layout, Home tiles, and entries in Users !' Screen Access
  - New requirePageAccess middleware lets granted non-admins use the announcements API
- Settings language selector no longer jumps the page scroll

File	+Lines	" Lines	Note

electron/main.js	+42	" 3	Single-instance lock, autoplay, media-key capture off
src/backend/middleware/auth.js	+16	" 1	requirePageAccess()
src/backend/routes/announcements.js	+36		Page-access gated management routes
src/frontend/src/pages/ScreenShell.jsx	+37		NEW — standalone screen wrapper
src/frontend/src/App.jsx	+13		Standalone routes

v1.5.46 MINOR UPDATE

04 Jul 2026
commit 311fcb2

Start-a-New-Day Archival, Automatic Backups, Offline License Activation

Files changed: 23 · +634 " 20 lines

- tickets.archived column (migration 005) — live queues and display hide archived rows, reports still read everything
- Settings !' "Start a New Day" archives current tickets instead of deleting them
- Automatic daily online DB backup, last 14 kept (database/backup.js)
  - Settings !' Backups: Backup Now, list, Open Folder, Restore (staged file swapped in on next start)
- Offline Ed25519 license verification — Server build shows an activation screen, key bound to machine id
  - tools/generate-license.js mints per-school keys; private key kept in .keys/ (gitignored, never shipped)

File	+Lines	" Lines	Note
src/backend/database/backup.js	+54		NEW — online backup + prune
electron/license.js	+87		NEW — Ed25519 verification
electron/activation.html	+84		NEW — activation screen
src/backend/routes/admin.js	+39		archive-tickets, backup endpoints
src/frontend/src/pages/admin/Settings.jsx	+128		New Day + Backups sections

v1.5.47 PATCH

12 Oct 2026
commit fcb4798

Merged the Unpushed v1.5.36 Work into the v1.5.46 Line

Files changed: 8 · +367 " 11 lines

- v1.5.36 (version badge + changelog) was committed locally after GitHub had moved to v1.5.46
- Merge commit joins both histories — nothing from either side was lost; fast-forward push, no force push

v1.5.48 BUG FIX

12 Oct 2026
commit this release

Correct Service Times & Clock Display, No Duplicate Ticket Numbers After Skip

Files changed: 14 · +161 " 28 lines

- Service time was inflated by the server's UTC offset (+180 min in Jordan) — SQLite UTC timestamps were parsed as local time
  - Migration 006 recomputes existing service_duration values from called_at / completed_at
- Reception error "UNIQUE constraint failed" after a skip — numbering now continues from the highest sequence issued today
- Printed ticket, staff dashboard wait time, reports, audit log, users list showed times 3 hours early — fixed
- Complete is refused unless the ticket is currently being served (no completing waiting or finished tickets)
- Call-next and transfer record the logged-in user, not a staff_id sent by the browser
- 10 new unit tests (queue.test.js); verified with a full API/socket and browser end-to-end run

File	+Lines	" Lines	Note
src/backend/controllers/queueController.js	+6	" 3	UTC parse, status guard, req.user
src/backend/controllers/ticketController.js	+10	" 11	Max-sequence numbering

src/backend/database/migrations/runner.js	+9		Migration 006
src/backend/tests/queue.test.js	+122		NEW — 10 tests
src/frontend/src/ (6 files)	+10	" 10	UTC-correct time display

v1.5.49 SECURITY

12 Oct 2026
commit this release

License Signing Keypair Replaced

Files changed: 7 · +1 " 1lines

- The original private signing key was not available on the build machine, so a new Ed25519 keypair was generated
- electron/license.js now embeds the new public key — license keys minted with the old key no longer activate on 1.5.49+
  - Already-activated installs on an older version are unaffected until they upgrade; then they need a key from the new private key
- New private key stored in .keys/license-private.pem (gitignored, never shipped) — back it up offline

File	+Lines	" Lines	Note
electron/license.js	+1	" 1	New PUBLIC_KEY

v1.5.50 BUG FIX

13 Oct 2026
commit this release

First-Time Setup "Test Connection" No Longer Fails Against a Running Server

Files changed: 1 · +6 " 8lines

- Test Connection always reported "Could not reach the server", even though Save & Launch connected fine
- Cause: the setup page is a file:// page (origin "null"); its fetch() was blocked by the server's CORS rules
- Fix: test via the existing main-process testServer() IPC (Node http — no CORS), same check ServerSetup.jsx uses

File	+Lines	" Lines	Note
electron/setup.html	+6	" 8	testConnection() uses electronAPI.testServer

v1.6.0 MINOR UPDATE

13 Oct 2026
commit this release

Staff PCs Find the Server Automatically — Even When It Moved Port

Files changed: 12 · +357 " 61lines

- Server answers a UDP discovery probe (port 41234) with school name, version and the HTTP port it actually bound
- Staff setup + in-app Server Settings: "Find Server on Network" — lists servers, click to use; auto-searches on first run
  - One server found !' selected automatically; manual entry accepts IP:port (e.g. 192.168.1.5:3001)
- Staff startup self-heals: if the saved server does not answer as SchoolQ, it searches the LAN and saves the new address
- Connection checks now verify the reply is SchoolQ — before, any program listening on the port passed
  - With another app on 3000 and SchoolQ on 3001, Staff used to load the other app's page
- Fixed: after a port retry, stale listen() callbacks also fired — the server announced the busy port
- All ports 3000–3010 busy: clear error dialog instead of the app silently closing

File	+Lines	" Lines	Note
electron/discovery.js	+105		NEW — discover, testServer, findServer
src/backend/discovery.js	+26		NEW — UDP responder
src/backend/server.js	+25	" 9	Single listening handler, responder, ports-busy
electron/main.js	+21	" 38	Self-healing Staff startup, IPC
electron/setup.html	+77	" 10	Find Server UI, IP:port
src/frontend/src/pages/ServerSetup.jsx	+45		Find Server UI

src/backend/tests/discovery.test.js	+53	NEW — 4 real-UDP tests

v1.7.0 MINOR UPDATE

13 Oct 2026
commit this release

Ticket Calls in Arabic & English With No Internet — Pre-Recorded Neural Voice

Files changed: 12 · +530 " 23lines

- Problem: Arabic calls needed internet (Google TTS); offline fallback had no Arabic voice — calls were silent or garbled
- Calls are now assembled from recorded clips: «b¦Fb†Jdr Edb AcfDd2à bvDb¦0d61b' 1d&E» + «dVJb›² ² ¶HbæEc6) d†9cF1d¦F» + «bvDd&3d[²
  - Voice: VoiceTut-TTS "Sarah" (Egyptian Arabic + English, Apache-2.0), generated once on the developer PC
  - 315 number clips ship with the app (1–999 in both languages, 3.7 MB); Arabic uses the «d‹² `orm after hundreds
  - Per-school clips (template text + department names) live in %APPDATA%\SchoolQ\voice — survive updates
- Schools keep their own call templates: each template is split into text pieces and {ticket}/{department}/{room}
- Any piece without a recording falls back to the online voice for that piece only — a call is never silent
- Settings !' Voice: Recorded (offline) / Online, with a list of pieces that still have no recording
- Clips preload while earlier ones play and are chained on "ended" — no overlap, tight joins
- tools/voice/generate_voice_pack.py (dev-only): numbers + per-school packs, resumable, silence-trim + normalize
  - Fixed during generation: digits glued to «d‹² pere skipped by the normalizer — all 99 «d‹² 6Æ— 2 &VvVæW ated
- Verified: 5 voice-assembly unit tests; E2E — ticket 125, both languages, 14 clips in order, zero online TTS

File	+Lines	" Lines	Note
tools/voice/generate_voice_pack.py	+174		NEW — VoiceTut clip generator
src/frontend/src/lib/voiceClips.js	+91		NEW — template split, number clips, plan
src/frontend/src/pages/PublicDisplay.jsx	+44	" 17	Recorded playback + per-piece fallback
src/frontend/src/pages/admin/Settings.jsx	+64	" 1	Voice mode + missing-recordings list
src/backend/routes/voice.js	+24		NEW — per-school clips + manifest
src/frontend/src/lib/voiceClips.test.js	+63		NEW — 5 tests (node --test)
src/frontend/public/voice/ (315 clips)			Numbers 1–999, AR + EN, 3.7 MB

v1.8.0 MINOR UPDATE

14 Oct 2026
commit this release

Record Any Phrase on the School Server in Advance; Announce on the Display Offline

Files changed: 20 · +842 " 40lines

- Optional Voice Engine add-on (~5 GB): VoiceTut model + private Python on the server PC, one-click install
  - install-engine.ps1: resumable downloads with SHA-256 checks, NVIDIA GPU !' CUDA build, otherwise CPU
  - Measured: ~1 min per phrase on a GPU (Quadro M2000); 3–7 min per phrase on an old 8-core CPU
- Admin !' Voice Recordings: every piece (calls, departments, announcements, own phrases) with status
  - Generate all missing · Record any phrase in advance · %¶ play · Ø=Ý re-record · Ø=ÝÑ delete
- Engine runs only when asked, never during a call: model loads once, stays 10 min, then frees memory
- New recordings reach displays instantly (voice_updated socket event) — no restart, no 30 s wait
- Announcements: Ø=Üâ "Announce on display" plays the message on the TV, recorded voice when available
- Shared voicegen.py: the dev tool and the server engine produce identical clips
- Verified: 8 engine tests (fake worker), 2 new voice tests; E2E with the real engine — record !' Ø=Üâ !' plays offline (10/10)

File	+Lines	" Lines	Note
src/backend/voice-engine/engine.js	+183		NEW — queue, worker lifecycle, manifest
src/backend/voice-engine/worker.py	+58		NEW — model stays loaded, JSON-lines
src/backend/voice-engine/voicegen.py	+58		NEW — shared trim/normalize/synthesis
src/backend/voice-engine/install-engine.ps1	+115		NEW — add-on installer
src/frontend/src/pages/admin/VoiceRecordings.jsx	+237		NEW — admin page
src/backend/routes/voice.js	+56		engine, generate, remove, install, announce

src/backend/tests/voiceEngine.test.js	+96	NEW — 8 tests

v1.9.0 MINOR UPDATE

14 Oct 2026
commit this release

Pick Any of the 17 VoiceTut Voices — Prepared in Advance, Never Mixed

Files changed: 12 · +330 " 90lines

- Voice Recordings !' Voice: all 17 voices (male + female) with a %¶ sample of each
- "Prepare" records a voice's 315 number clips + all phrases in the background (resumable)
  - ~5 h per voice with a GPU — Sarah ships ready, so she never needs preparing
- "Use" switches every announcement to that voice — only once fully prepared, so calls never mix voices
- Each voice keeps its own recordings (voice/<Name>/) — switching back and forth is instant
- Al-Noor 1.7.0 recordings move automatically into voice/Sarah/ on first start
- Number-word logic now lives once in voicegen.py (dev tool + engine share it)
- Verified: 13 engine tests, 8 voice tests; E2E with the real engine — prepare Yasmin, switch, display plays her clips (14/14)

File	+Lines	" Lines	Note
src/backend/voice-engine/engine.js	+90	" 40	Per-voice folders, number jobs, migration
src/backend/routes/voice.js	+70	" 15	voices, sample, prepare, use
src/backend/voice-engine/voicegen.py	+45		number_slots / number_text
src/frontend/src/pages/admin/VoiceRecordings.jsx	+95	" 10	Voice picker
src/backend/tests/voiceEngine.test.js	+60	" 15	13 tests (+5)

v1.10.0 MINOR UPDATE

15 Oct 2026
commit this release

Separate Call & Announcement Voices, Pause Only Playing Media, Banner Over Video, Instant Updates

Files changed: 20 · +520 " 110lines

- Two voices: "Use for calls" (needs the 315 number clips) and "Use for announcements" (any voice, right away)
  - Choosing an announcement voice records your announcement sentences in it automatically
- Voice engine: #ø Pause / %¶ Resume — the clip in progress finishes, the GPU is freed while paused
- Priority queue: a single recording jumps ahead of a voice preparation (minutes instead of hours)
- Smart media pause: asks Windows what is actually playing, pauses only that, resumes only what SchoolQ paused
  - Before: a blind Play/Pause key press — with nothing playing it could START music
- Call banner: ticket, department (EN + AR) and room appear over any video / app during the call, then fade
  - Never takes focus (video stays fullscreen), click-through; setting: only when media plays / always / never
- Display refreshes instantly on every queue change, settings change and announcement change (timers kept as backup)
  - Fixed: new announcements only reached the ticker after the display was reloaded
- %¶ preview on the Announcements page plays the real recording when there is one
- Fixed: the hidden banner window would have kept the app running after closing SchoolQ
- Verified: 16 engine tests, banner window check (8), display E2E with media/banner/two voices/instant updates (13/13), API regression 53/53

File	+Lines	" Lines	Note
electron/mediaControl.js	+105		NEW — Windows media sessions helper
electron/banner.js + banner.html	+170		NEW — call banner window
electron/main.js	+20	" 22	media/banner IPC, old media key removed
src/backend/voice-engine/engine.js	+35	" 6	pause/resume, priority
src/backend/routes/voice.js	+40	" 12	announcement voice, pause/resume
src/frontend/src/pages/PublicDisplay.jsx	+60	" 35	media + banner flow, live events
src/frontend/src/pages/admin/VoiceRecordings.jsx	+70	" 25	two voices, pause button

Project Statistics

Cumulative metrics across all versions

VERSION TIMELINE

v1.2.0	06 May 2026	Initial release — 73 files, 16,712 lines of code
v1.3.0	06 May 2026	Bilingual TTS fix — AbortController, 700ms pause, MP3 validation
v1.3.1	07 May 2026	CORS restriction, rate limiting, settings cache, compression
v1.3.2	07 May 2026	JWT 8h!'24h, 401 auto-logout interceptor
v1.3.3	07 May 2026	No-show setting hardcoded!'DB-driven
v1.3.4	07 May 2026	Electron startup silent crash !' error dialog
v1.3.5	07 May 2026	Native module ABI mismatch — buildDependenciesFromSource
v1.3.6	07 May 2026	Final call template — respects no_show_after_calls
v1.4.0	07 May 2026	Screen permissions, logo upload, audit log, menu bar removed
v1.5.15	15 May 2026	Service types × 3 features, room number, sequential TTS fix
v1.5.16	20 May 2026	Force logout, socket JWT auth, is_active check, JWT!'8h
v1.5.17	20 May 2026	Input validation, per-install secret, rate limit, error handler, indexes
v1.5.18	20 May 2026	Toast notifications, skeleton loader, migration runner, 16 tests
v1.5.25	02 Jun 2026	Cancel/Left action, no-show reason, granular report permissions, ticket log wait fix
v1.5.26	03 Jun 2026	Reset All Ticket Data — Danger Zone in admin settings
v1.5.27	03 Jun 2026	Self-service kiosk — touch panel for parents to get tickets without staff
v1.5.28	03 Jun 2026	Auto port scan 3000–3010, no longer kills foreign processes
v1.5.29	03 Jun 2026	Temporal dead zone crash fix in ticketController exports
v1.5.30	03 Jun 2026	Kiosk auto-prints ticket + kiosk permission in screen access
v1.5.31	03 Jun 2026	Export/import users & settings, scrollbar fix, migrations restored
v1.5.32	03 Jun 2026	db.exec() for DDL in migration runner, stack trace in error dialog
v1.5.33	03 Jun 2026	Migrations table schema mismatch — PRAGMA detect and recreate
v1.5.34	03 Jun 2026	Dynamic port in ServerSetup — test uses real IP:port, not hardcoded 3000
v1.5.35	03 Jun 2026	Media key pause/resume — music stops during TTS announcements
v1.5.36	13 Jul 2026	Version badge in admin settings header, Vite __APP_VERSION__ injection
v1.5.45	04 Jul 2026	Single-instance server, announcement audio fixes, standalone screens
v1.5.46	04 Jul 2026	New-day archival, automatic backups, offline license activation
v1.5.47	12 Oct 2026	Merge of unpushed v1.5.36 work into the v1.5.46 line

v1.5.48

12 Oct 2026

		Timezone-correct service times & clocks, no duplicate numbers after skip
v1.5.49	12 Oct 2026	New license signing keypair (old private key unavailable)
v1.5.50	13 Oct 2026	Staff first-time setup: Test Connection works (CORS-free IPC check)
v1.6.0	13 Oct 2026	LAN server discovery, self-healing Staff startup, port-conflict handling
v1.7.0	13 Oct 2026	Offline recorded voice: Arabic + English calls from clips, no internet
v1.8.0	14 Oct 2026	Voice Engine on the server, Voice Recordings page, spoken announcements
v1.9.0	14 Oct 2026	Choose any of 17 voices — prepare in advance, switch instantly
v1.10.0	15 Oct 2026	Two voices, pause engine, smart media pause, call banner, instant display

CUMULATIVE CHANGE STATISTICS

Total versions released: 37 (v1.2.0 !' v1.10.0)
Total git commits: 25+ commits on main branch
Total files ever modified: ~80 unique files touched
Lines added across all commits: ~22,000+
Lines removed across all commits: ~2,200+
Backend files modified: ~20 (routes, controllers, middleware, DB)
Frontend files modified: ~30 (pages, components, stores, hooks)
New files created: 13 (audit.js, settingsCache.js, validate.js, errorHandler.js,

runner.js, Toast.jsx, useToastStore.js, 2 new pages, 3 test files, scripts/)
Files deleted: 2 (AdminPanel.jsx, DisplayMonitor.jsx — legacy)
Test coverage: 54 tests — auth, validation, queue logic, LAN discovery, offline voice, voice engine, voices
Supported platforms: Windows 10/11 x64 (Electron 41)
Node.js runtime

(backend): Electron bundled Node (v24-based, ABI 145)

MATURITY ASSESSMENT BY AREA

B+	**Authentication & Sessions**	JWT, bcrypt, force-logout, is_active check, socket auth, 8h expiry
B	**Input Validation**	express-validator on all write endpoints, 422 with clear messages
B	**Authorization**	Role + allowed_pages system, ProtectedRoute, requireRole middleware
B	**Rate Limiting**	Login limiter (20/15min) + mutation limiter (60/min) on all writes
B-	**Error Handling**	Centralized handler, no stack trace exposure, toast UI; no retry logic
B	**Database Design**	WAL, foreign keys, 10 indexes, migration runner, transactions
A-	**Real-time Features**	Socket.io with auth, force-logout events, dept rooms, user tracking
C+	**Logging & Audit**	Audit log table + morgan; no structured logging, no rotation
C+	**Testing**	54 unit tests; E2E run manually, not automated; no CI pipeline
B+	**Code Organization**	Clean route/controller split, Zustand, API layer, no fat controllers
B	**User Experience**	Toast system, skeleton loaders, real-time updates, bilingual voice
B-	**Build & Deployment**	2 installer targets, per-install secret; no auto-update, no code signing

Overall: Mid-Level — Production-Ready for Controlled School Environment

SchoolQ is a well-architected, fully functional system. Security, real-time features, and UX are solid. Remaining gaps (structured logging, E2E tests, CI/CD) are optional for a single-school deployment but recommended before wider rollout.